Paper 10 - T10 Master Receipt

CQE-paper-10
CQECMPLX Formal Paper Series
Review-facing PDF built 2026-06-12

Construction Basis

Bind Paper 00 and Papers 01-09 into a replayable master receipt carried from the observer's step-00 enumeration event into step 1, with typed transport rows, materialized lookup receipts, and visible open-lift boundaries.

This PDF is the clean paper artifact. Source extracts, kernels, receipts, tool notes, workbook material, and package bindings are used as construction evidence; they are not treated as separate review-facing papers.

Evidence Inputs

- promoted formal paper: production\formal-papers\CQE-paper-10\FORMAL_PAPER.md
- paper kernel manifest: production\paper-kernels\papers\CQE-paper-10\paper_kernel_manifest.json
- verifier: production\formal-papers\CQE-paper-10\verify_t10_master_receipt.py
- receipt: production\formal-papers\CQE-paper-10\t10_master_receipt.json (T10 master receipt integrity: pass)
- body: production\papers\CQE-paper-10\PAPER-BODY.md
- source: production\papers\CQE-paper-10\SOURCE.md
- formal: production\papers\CQE-paper-10\01-CQE-formal\FORMAL.md
- tool: production\papers\CQE-paper-10\02-CQE-tool\TOOL.md
- workbook: production\papers\CQE-paper-10\03-CQE-workbook\WORKBOOK.md
- recovered_output: production\lib-forge\recovered\papers_output\CQE-paper-10.md

Paper 10 - T10 Master Receipt

Abstract

Paper 10 formalizes the first stack-level receipt in the CQECMPLX sequence. Its object is not a new physical claim; it is the proof that Papers 00-09 are bound into one inspectable and replayable unit. The master receipt verifies three layers at once: Paper 00 as the inherited information-burden contract and initial observer enumeration event, Papers 01-09 as promoted receipt-bearing formal papers, and the transport and lookup substrate that records what is demonstrated, what is locally bounded, and what remains an open lift.

The theorem is closed for receipt integrity. It proves that the 00-09 substack has source bindings, formal receipts, typed transport rows, replayable local witnesses, and a materialized lookup cache. It does not claim that every registered lift is already demonstrated. The honest verdict is therefore a passing master receipt with visible open-lift boundaries.

Proof/Exposure Hierarchy

The proof-carrying content of this paper is the mathematics: the definitions, receipt equations, transport classifications, cache materialization checks, and replayable verifier. Paper 00, workbook routes, analog tools, and open obligation ledgers are supplemental validation and exposure layers. They make the receipt inspectable by hand or by software, but they are not the primary result. The primary result is the master-receipt theorem below.

Definitions

A **paper receipt binding** is a pair `(paper_id, receipt_path)` such that the receipt exists, can be parsed, and reports a pass-like status for the theorem it carries.

The **observer center `C`** is the active center introduced by a requested enumeration event. It is not a passive label. It is the fact that an observer has asked the system to enumerate something, and the system has encoded that request as the center against which later left/right, boundary, transform, residue, and receipt states are read.

The **step `00 -> 1` encoding event** is the transition from the inherited Paper 00 burden contract into the first active paper. Paper 00 defines what must be carried; Paper 01 begins carrying it. Every later receipt is an effect of that initial choice unless a later paper explicitly introduces a new enumeration center and proves the handoff.

A **transport obligation row** is a typed record:

```
(id, source_object, target_object, map, preserved_quantity,
 failure_condition, witness, classification, proof_boundary)
```

The allowed classifications are:

```
demonstrated
bounded_local
bounded_external
registered_landing_forms
open
```

A **lookup receipt** is:

```
(kind, key, value, source_id, evidence_level, complexity_claim)
```

A **T10 master receipt** is the tuple:

```
T10 = (C00, E00->1, P00, P01..P09, R, L, V, O)
```

where `C00` is the observer-bound enumeration center, `E00->1` is the initial encoding event from Paper 00 into Paper 01, `P00` is the Paper 00 contract binding, `P01..P09` are formal paper receipt bindings, `R` is the transport obligation table, `L` is the materialized lookup cache, `V` is the verifier verdict, and `O` is the visible open-lift set.

Main Claim

Theorem 10.1, T10 Master Receipt Integrity. The CQECMPLX substack consisting of Paper 00 and Papers 01-09 is inspectable and replayable as a single receipt object. The receipt proves:

```
1. Paper 00 is bound as the inherited minimum information contract and
   observer enumeration event.
2. Papers 01-09 have promoted formal receipts with pass-like status.
3. The four transport rows have required fields and valid classifications.
4. The local witnesses replay.
5. Two transport rows are demonstrated and two remain visible non-demonstrated lifts.
6. The lookup cache materializes the one-million-bit Rule 30 window, 157
   unipotent orbits, 24 lattice forms, and the UMRK/LMFDB source registers.
7. The Prize 3 lookup substrate keeps `closed_form_claim = False`, so the
   receipt does not overclaim cold-start closure.
```

Proof

Bind Paper 00 by requiring both its source contract and at least one Paper 00 proof receipt. This establishes the inherited contract layer without treating Paper 00 as one of the active future papers. It also establishes `C00`: the observer's requested enumeration encoded as the system center. The transition from Paper 00 to Paper 01 is therefore not merely editorial order. It is the first active encoding event:

```
observer request -> C00 -> E00->1 -> first carried paper state
```

Every later paper in the bound substack is read as a transported consequence of that event unless it explicitly proves a recentering.

For each `i` from `1` through `9`, bind `CQE-paper-i` to its promoted formal receipt. Each binding must exist and report a pass-like status. These bindings form the paper component of `T10`.

Next construct the transport table `R` using the four registered rows:

```
LCR_TO_D4_AXIS_SHEET
D4_TO_J3O_DIAGONAL_CARRIER
J3O_TO_G2_F4_T5A_ROUTE
EXCEPTIONAL_ROUTE_TO_NIEMEIER_LANDING_FORMS
```

The verifier checks that every row has the required fields and that every classification belongs to the allowed classification set. It then replays the local witnesses:

```
verify_chart_codec_d4
verify_j3o_axioms
verify_conjugate_triple
verify_niemeier_landing_registry
```

The resulting transport verdict is `pass_with_open_lifts`. This is a proof of inspectability, not a disguised claim that all lifts are closed. The table contains two demonstrated rows and two open or registered/bounded lift rows. Therefore the open boundary is preserved as part of the proof object.

Finally materialize the lookup cache `L`. The cache must expose:

```
rule30_bits = 1000000
unipotent_orbits = 157
lattice_forms = 24
source_registers.umrk = true
source_registers.lmfdb = true
```

The verifier also reads bit `999999` as a `LookupReceipt` from `wolfram-rule30-center-million` and constructs a Prize 3 lookup receipt at `N=4096, group=F4`. That receipt is accepted only because it keeps `closed_form_claim = False` and names the remaining obligation to prove a cold-start `N`-to-axis/sheet or `N`-to-Weyl-fingerprint map.

All components of `T10` are therefore present, typed, replayable, and honest about their boundaries. QED.

Relation to Earlier Papers

Papers 01-09 build the first carrier chain after the observer's enumeration event has been encoded: LCR carrier, correction surface, triality surface, boundary repair, oloid path carrier, causal code, discrete/continuous bridge, lattice closure template, and Hamiltonian window emergence. Paper 10 wraps them as a receipt object:

```
observer request at Paper 00
-> C00
-> 00-to-1 encoding event
-> paper receipts
-> transport rows
-> local witness replay
-> lookup receipts
-> pass verdict with visible open lifts
```

This is why Paper 10 belongs at the start of the second block. It converts the first block and its immediate temporal extension into a stack-level audit object that later papers can cite.

Falsifier

The verifier rejects these overclaims:

```
"T10 proves every registered lift is already demonstrated"
"The lookup cache makes a cold-start closed-form N-to-fingerprint claim"
"A paper enters the master receipt without a source or receipt binding"
"A later paper can ignore the observer enumeration event encoded at 00 -> 1"
```

The paper passes because it proves receipt integrity while refusing to erase open obligations.

Code Reconstruction

The production verifier is:

```
production/formal-papers/CQE-paper-10/verify_t10_master_receipt.py
```

It emits:

```
production/formal-papers/CQE-paper-10/t10_master_receipt.json
```

The verifier checks:

```
1. Paper 00 source and proof-receipt binding.
2. Paper 00 as observer enumeration contract and `00 -> 1` encoding event.
3. Paper 01-09 promoted formal receipt bindings.
4. Transport row schema, classification validity, and local witness replay.
5. Demonstrated/open lift counts: 2 demonstrated and 2 visible non-demonstrated lifts.
6. Lookup cache materialization against the packaged source datasets.
7. Prize 3 lookup honesty boundary: no cold-start closed-form claim.
```

Open Audit Boundaries

- Paper 10 does not close the cold-start `N`-to-axis/sheet map.
- Paper 10 does not close the `N`-to-Weyl-fingerprint map.
- Registered Niemeier landing forms are valid receipt targets, not automatic proof closure.
- Later papers may cite T10 as a master receipt, but they must still prove their own domain claims.
- A later recentering is allowed only when it explicitly records the new observer/enumeration event and its handoff from the prior center.

Conclusion

Paper 10 proves that the first CQECMPLX substack is not a pile of adjacent claims; it is a single replayable master receipt carried from the observer's initial enumeration event. The result is proof-first: `C00`, the `00 -> 1` encoding event, receipt bindings, typed transport rows, local witnesses, lookup receipts, and explicit open boundaries are all present in one verifiable object.